Causal Proof of Reasoning: Structural Auditing of Latent Dependencies for Defending Federated Learning

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***Abstract*—Federated Learning (FL) enables collaborative model training across distributed, privacy-sensitive participants. However, its decentralised structure creates a severe vulnerability: malicious clients can inject backdoor triggers or poison the global model's reasoning logic without detection by conventional statistical defenses. Existing Byzantine-robust aggregators analyse weight-vector divergence and therefore fail against semantic attacks, where an adversary's model behaves correctly on clean data while harbouring a dormant trigger—a characteristic that preserves statistical similarity to honest updates. We propose Causal Proof of Reasoning (PoR), a structural auditing framework that shifts the defense layer from numerical weight analysis to topological inspection of latent feature dependencies. In PoR, each client applies the NOTEARS continuous optimisation algorithm to its latent representation layer, extracting a Directed Acyclic Graph (DAG) encoding the causal structure of its learned features. A server-side *Logic Validator*, implemented as a Siamese Graph Neural Network (SimGNN), compares each submitted DAG against a maintained global consensus graph, approximating Graph Edit Distance (GED) in $\mathcal{O}(1)$ time. Honest clients produce DAGs topologically close to the consensus; poisoned clients exhibit structural deformation, evidenced empirically by a mean Wasserstein distance of 5.25 across latent dimensions between clean and poisoned representations. PoR is deployed within FedNEAT, a Federated NeuroEvolution framework, demonstrating that the defense holds under structurally heterogeneous client models. Extensive evaluation on the ASIA (8-node) and ALARM (37-node) Bayesian network benchmarks shows that PoR reduces Attack Success Rate (ASR) from 60.30% to 0.40% on ALARM and from 5.62% to 0.52% on ASIA, while maintaining model utility.**

***Index Terms*—Federated learning, backdoor defense, causal discovery, graph neural networks, Byzantine robustness, structural auditing, NOTEARS, neuroevolution.**

I. Introduction

Federated Learning (FL) [1] has emerged as a foundational paradigm for training machine learning models across distributed, data-heterogeneous participants without centralising raw data. By aggregating only model updates rather than datasets, FL offers strong privacy guarantees and enables collaboration across healthcare, finance, and autonomous systems—domains where data governance prohibits centralised collection [2], [3].

Despite these advantages, FL's distributed trust model exposes it to a fundamental class of threats: *model poisoning and backdoor attacks*. A malicious participant can submit crafted model updates that embed a hidden trigger into the global model, causing it to misclassify any input containing that trigger while maintaining normal accuracy on clean inputs [4], [5]. Unlike data poisoning, model poisoning attacks operate entirely within the update protocol—the adversary's submission is indistinguishable from legitimate updates at the parameter level. More sophisticated Distributed Backdoor Attacks (DBA) [6] decompose a global trigger across multiple malicious agents, each submitting a seemingly benign partial pattern that assembles into a potent trigger only at the global model, yielding Attack Success Rates (ASR) exceeding 85% even against state-of-the-art robust aggregators.

A. The Insufficiency of Weight-Space Defenses

The dominant family of Byzantine-robust aggregators—including Krum [7], Trimmed Mean, and Median [8]—operates by computing pairwise distances or statistics over the high-dimensional weight vectors submitted by clients, discarding statistical outliers. These methods embed a critical assumption: *a malicious update must be statistically anomalous in weight space*. This assumption is routinely violated.

An adversary who scales their poisoned update to match the ℓ_2-norm of honest updates, or who mounts a DBA with sub-triggers that individually appear within the honest distribution, will evade all weight-divergence detectors. Even FLTrust [9], which bootstraps trust from a server-held root dataset, is susceptible when the trigger is semantically aligned with legitimate feature correlations. A parallel vulnerability is exploited at the explainability layer: Scaffolding attacks [10], [11] instruct the adversary's model to switch between a malicious inference function and a benign explanation function depending on whether the input is in-distribution, thereby satisfying post-hoc auditing tools such as LIME and SHAP while remaining malicious in deployment.

The core failure mode is conceptual: *these defenses audit what a model predicts under specific inputs, not how it reasons internally*. An adversary who preserves the statistical signature of their weight updates while corrupting the underlying reasoning structure will evade detection indefinitely.

B. Causal Proof of Reasoning

We propose a fundamentally different auditing paradigm. Rather than inspecting weight magnitudes or output distributions, we inspect the *causal dependency structure* of a client's latent feature representations—the topology of relationships between learned concepts that governs the client's reasoning. A model that has been poisoned must corrupt its internal feature dependencies to associate the trigger with the target class; this corruption is reflected as a topological deformation of the latent Directed Acyclic Graph (DAG) relative to the honest global consensus.

Our framework, **Causal Proof of Reasoning (PoR)**, operationalises this intuition as follows. Each client applies the NOTEARS algorithm—a continuous optimisation procedure that enforces strict DAG constraints [12]—to its latent representation layer, producing a compact DAG as a structural certificate of its reasoning. The server maintains a *Global Consensus Graph*—a momentum-updated aggregate of past honest client structures—and employs a pre-trained SimGNN [13] to predict the Graph Edit Distance (GED) between each submitted DAG and the consensus in $\mathcal{O}(1)$ inference time, regardless of graph size. Clients whose structural certificates deviate significantly from the consensus are identified as adversarial and excluded from aggregation.

PoR is deployed within FedNEAT: a Federated NeuroEvolution of Augmenting Topologies framework, where clients evolve heterogeneous neural architectures via genetic mutation rather than fixed-architecture gradient descent. FedNEAT deliberately eliminates the homogeneous gradient signature that statistical defenses rely on, making it the most adversarially challenging substrate in which to evaluate PoR. The system is implemented using the Flower federated learning framework [14] with Ray-based parallelism.

C. Contributions

The primary contributions of this work are:

1) **The PoR Framework**: A novel structural auditing mechanism for FL that extracts and compares latent DAGs to detect poisoned clients, operating orthogonally to weight-space defenses.
2) $\mathcal{O}(1)$ **Logic Validation via SimGNN**: A Siamese GNN that approximates GED over arbitrary Bayesian network topologies in constant inference time, enabling real-time per-round validation at federation scale.
3) **Dynamic Percentile Thresholding**: An adaptive rejection boundary derived from the expected Byzantine ratio, preventing honest stragglers from being rejected under non-IID data drift.
4) **Empirical Validation on Bayesian Benchmarks**: Comprehensive evaluation on ASIA and ALARM Bayesian networks, demonstrating ASR suppression from 60.30% to 0.40% on ALARM and from 5.62% to 0.52% on ASIA.
5) **Latent Deformation Analysis**: Empirical evidence via Wasserstein distance ($\bar{W}_1 = 5.25$) that backdoor poisoning produces statistically distinguishable latent manifold deformation, providing the theoretical justification for structural auditing.

The remainder of this paper is organised as follows. Section II surveys related work. Section III formalises the federated system and threat model. Section IV presents the PoR methodology. Section V describes the experimental setup. Section VI presents results and analysis. Section VII discusses implications and limitations. Section VIII concludes.

II. Related Work

A. Byzantine-Robust Federated Aggregation

The foundational challenge of FL under Byzantine faults was formalised by Blanchard et al. [7], who proposed *Krum*—a server-side aggregator that selects the client update with the minimum sum of distances to its nearest neighbours, provably tolerating up to $f < n/2$ Byzantine clients. Yin et al. [8] established statistical optimality bounds for coordinate-wise Trimmed Mean and Median aggregators. Despite theoretical guarantees, these methods share the assumption that adversarial updates are statistical outliers in weight space—an assumption systematically violated by norm-constrained and DBA-style attacks.

FLTrust [9] addresses this by maintaining a server-held root dataset to compute trust scores for each client's update direction, gating aggregation by cosine similarity. Our baseline in this work— *Byzantine-Robust FedAvg*—implements a similar cosine weight-divergence filter, representing the current practical state of the art for deployed systems. FLAME [15] combines clustering with adaptive noise injection but introduces significant utility degradation. None of these approaches inspect the *internal reasoning structure* of client models.

B. Backdoor and Poisoning Attacks

Backdoor attacks in FL were systematically studied by Bagdasaryan et al. [4], who showed that a single malicious client can dominate the global model's behaviour through *model replacement*—scaling their poisoned update by $1/q$ where q is the learning rate. Chen et al. [5] demonstrated trigger-based targeted misclassification in centralised settings. Xie et al. [6] introduced Distributed Backdoor Attacks (DBA), decomposing a global trigger across multiple participants to evade per-client anomaly detection. Sun et al. [16] provided a theoretical analysis of backdoor durability, showing that triggers persist across rounds of aggregation. Wang et al. [17] demonstrated that edge-case backdoors, targeting tail distributions, survive robust aggregation.

C. Explanation Poisoning and Scaffolding

A parallel attack surface exploits post-hoc explainability tools. Slack et al. [10] demonstrated that LIME and SHAP explanations can be systematically falsified by training an Out-of-Distribution (OOD) detector that switches model behaviour between inference and explanation modes. Heo et al. [11] showed that gradient-based saliency maps are similarly manipulable. These *scaffolding attacks* are particularly dangerous in FL, as a trusted model can present clean explanations to a central auditor while remaining malicious in deployment. PoR defends against scaffolding by auditing latent structure rather than output-space explanations, rendering the OOD-switching mechanism ineffective—structural poisoning cannot be hidden within the DAG.

D. Causal Discovery

Pearl's framework [18] established causal DAGs as the canonical language for encoding conditional independence relationships. Peters et al. [19] provided a rigorous treatment of identifiability conditions for causal structures from observational data. The NOTEARS algorithm [12] revolutionised computationally tractable causal structure learning by reformulating the combinatorial DAG constraint as a differentiable equality constraint, enabling gradient-based optimisation over the space of DAGs. We apply NOTEARS to latent representations rather than raw features, treating the learned representation space as the observational domain in which causal structure is extracted.

E. Graph Similarity and Neural Graph Matching

Exact GED computation is NP-hard [20], requiring exponential search in the worst case. SIMGNN [13] introduced a Siamese GNN architecture that learns to predict normalised GED directly, achieving approximately 1720× speedup over exact A* search on the AIDS dataset with MSE of 1.19×10^{-3}. The Message Passing Neural Network (MPNN) framework [21] and Graph Isomorphism Network (GIN) [22] provide the theoretical basis for the expressive power of GNN-based graph encoders. We build our Logic Validator on the SIMGNN architecture, pre-training it on structural perturbations of the consensus graph to learn the GED landscape of our specific Bayesian network topology.

F. NeuroEvolution in Federated Settings

Stanley and Miikkulainen's NEAT algorithm [23] introduced topology-evolving neural networks where both weights and connections evolve via genetic operators, producing heterogeneous architectures from a minimal initial structure. FEDNEAT extends this to the federated setting, where each client evolves a private genome and submits its structural update for aggregation rather than gradient vectors. This creates a fundamentally different threat surface: without homogeneous weight vectors, cosine-distance defenses are inapplicable, making POR's topology-based auditing the only viable defense strategy.

III. PROBLEM FORMULATION AND THREAT MODEL

A. Federated Learning System Model

Consider a federation of N clients $\mathcal{C} = \{c_1, c_2, \dots, c_N\}$ coordinated by a central server. Each client c_i holds a private local dataset $\mathcal{D}_i$ drawn from a distribution P_i over $\mathcal{X} \times \mathcal{Y}$, where $\mathcal{X} \subseteq \mathbb{R}^d$ is the feature space and $\mathcal{Y} = \{1, \dots, K\}$ is the label space. Datasets are non-IID: $P_i \neq P_j$ in general. The federation proceeds over T global rounds; in each round t, each client trains a local model on $\mathcal{D}_i$ and submits a structured update $U_i^{(t)}$ to the server, which aggregates across clients and broadcasts an updated global model $\theta^{(t)}$.

In FEDNEAT, updates $U_i^{(t)}$ are evolved genome structures rather than gradient vectors. Under POR, each update additionally includes a structural certificate $\mathcal{G}_i^{(t)}$—a DAG over the latent feature space.

B. Threat Model

We model a *Byzantine adversary* controlling a subset $\mathcal{B} \subset \mathcal{C}$ of $|\mathcal{B}| = m$ malicious clients, where $m < N/2$. The adversary has:

- **Local omniscience**: Full control over $\mathcal{D}_i$ for $c_i \in \mathcal{B}$, including data poisoning and model manipulation.
- **Protocol compliance**: The adversary submits updates that conform to the FL protocol. They do *not* compromise the server or honest clients' local data.
- **Trigger knowledge**: The adversary coordinates a shared backdoor trigger $\delta \in \mathcal{X}$ (or distributes sub-triggers across $\mathcal{B}$ as in DBA [6]).

Definition 1 (Backdoor Attack). *A backdoor attack succeeds if the global model $\theta^{(T)}$ satisfies:*

$$f_{\theta^{(T)}}(x) = y \quad \forall (x, y) \in \mathcal{D}_{clean} \tag{1}$$

$$f_{\theta^{(T)}}(x \oplus \delta) = y_t \quad \forall x \in \mathcal{X} \tag{2}$$

where $y_t \neq y$ is the adversary's target label and $\oplus$ denotes trigger injection into feature δ.

C. Poisoning Strategy

Each malicious client $c_i \in \mathcal{B}$ poisons a fraction ρ of its local training data by (i) setting a designated trigger feature $x_{[\delta]} \leftarrow 0$ and (ii) flipping the corresponding label to the target class y_t. This strategy is designed to maximise the probability that the model's latent representations associate the trigger feature with y_t while preserving accuracy on clean samples.

D. Evaluation Metrics

We evaluate defenses using two complementary metrics:

- **Attack Success Rate** (ASR): The fraction of triggered inputs misclassified as the target label by the poisoned global model. A lower ASR indicates a more effective defense.
- **Main Task Accuracy** (MTA): Accuracy on a clean held-out test set. Maintained MTA demonstrates the defense does not cause catastrophic utility loss for honest clients.

E. Defense Objective

A defense mechanism Φ is *effective* if, given a federation with m Byzantine clients, the protected global model $\theta_\Phi^{(T)}$ satisfies:

$$\mathrm{ASR}(\theta_\Phi^{(T)}) \ll \mathrm{ASR}(\theta_{\text{undefended}}^{(T)}) \tag{3}$$

$$\mathrm{MTA}(\theta_\Phi^{(T)}) \approx \mathrm{MTA}(\theta_{\text{honest}}^{(T)}) \tag{4}$$

POR aims to achieve this by replacing weight-space filtering with structural certificate validation.

IV. THE CAUSAL PROOF OF REASONING FRAMEWORK

Fig. 1 illustrates the end-to-end POR pipeline. Each federated round proceeds in four stages: (i) local genome evolution and feature extraction by clients, (ii) latent causal graph extraction via NOTEARS, (iii) server-side structural auditing via SIMGNN, and (iv) conditional aggregation of accepted updates only.

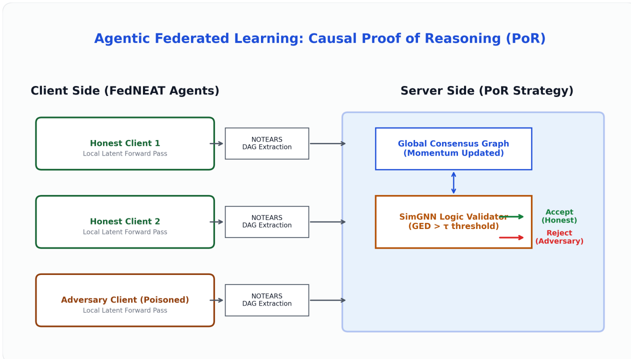

Fig. 1. The PoR federated round. Clients extract latent DAGs via NOTEARS and submit them alongside genome updates. The server Logic Validator (SimGNN) computes structural deviation from the Global Consensus Graph and rejects adversarial submissions before aggregation.

A. FedNEAT: Heterogeneous Client Architecture

Unlike conventional FL, where all clients train identical architectures via gradient descent, FedNEAT assigns each client a private *genome* Γ_i—a graph-encoded neural network topology that evolves through mutation across rounds. Following the NEAT formalism [23], each genome is a directed graph $\Gamma_i = (V_i, E_i)$ where nodes V_i represent neurons and edges E_i represent weighted synaptic connections tagged with globally unique *innovation numbers*.

In each round, each client c_i performs L local epochs of forward propagation on $\mathcal{D}_i$ using its current genome, computes a fitness score based on classification loss, and applies NEAT mutation operators: weight perturbation (probability $p_w = 0.8$), connection addition ($p_c = 0.1$), and node insertion ($p_n = 0.05$). The evolved genome is serialised and submitted to the server as the structural update $U_i^{(t)}$.

Critically, because genomes are heterogeneous—different clients may have different numbers of nodes and connections—there is no canonical weight-vector representation. This renders standard defenses based on cosine distance and Euclidean norms completely inapplicable, establishing FedNEAT as the most stringent possible testbed for PoR.

B. Latent Causal Graph Extraction via NOTEARS

After local training, each client extracts a DAG $\mathcal{G}_i$ from its learned latent representations. Let $Z_i \in \mathbb{R}^{n_i \times p}$ denote the matrix of p-dimensional latent activations collected over a local mini-batch of size n_i. We treat each column of Z_i as a random variable and seek a weighted adjacency matrix $W_i \in \mathbb{R}^{p \times p}$ encoding the conditional independence structure.

1) DAG Constraint: The NOTEARS algorithm [12] casts DAG structure learning as a continuous optimisation problem. The key insight is the algebraic characterisation of acyclicity:

Theorem 1 (Acyclicity Constraint [12])**.** *A matrix $W \in \mathbb{R}^{p\times p}$ is a DAG (i.e., has no directed cycles) if and only if*

$$h(W) \;=\; \mathrm{tr}\big(e^{W \circ W}\big) - p \;=\; 0 \tag{5}$$

where $\circ$ denotes the Hadamard (element-wise) product.

Each client solves the constrained optimisation:

$$\min_{W_i} \; \frac{1}{2n_i}\|Z_i - Z_i W_i\|_F^2 \;+\; \lambda \|W_i\|_1 \quad \text{s.t.} \quad h(W_i) = 0 \tag{6}$$

where the ℓ_1 penalty with coefficient λ enforces sparsity, promoting interpretable DAGs with few edges. Problem (6) is solved via augmented Lagrangian optimisation [12] for a maximum of $I_{\max}$ iterations.

2) Graph Construction: The resulting adjacency matrix W_i is thresholded at ϵ: edge (u, v) is retained if $|W_i[u, v]| > \epsilon$. The threshold ϵ is a global hyperparameter controlling DAG sparsity (default $\epsilon = 0.02$, tuned per dataset). The extracted DAG $\mathcal{G}_i = (V, E_i)$ shares a fixed node set V corresponding to the p latent dimensions and has a dataset-specific edge set E_i.

C. Latent Distribution Shift Analysis

To provide theoretical grounding for the structural auditing approach, we analyse how backdoor poisoning affects the latent feature distributions.

Proposition 1 (Latent Deformation Under Poisoning)**.** *Let $Z^{(\mathrm{h})}$ and $Z^{(\mathrm{adv})}$ denote the latent representations of an honest and an adversarial client respectively. Under a feature-zeroing backdoor attack with poison fraction ρ, the adversarial latent distribution is deformed with expected Wasserstein-1 distance:*

$$\mathbb{E}\big[W_1(Z^{(\mathrm{h})}, Z^{(\mathrm{adv})})\big] > 0 \tag{7}$$

as the trigger feature induces a systematic shift in the joint feature distribution that propagates through the latent layer.

We empirically measure this deformation by running matched honest and adversarial clients on the ASIA dataset and computing the per-dimension Wasserstein-1 distance between their latent activation distributions. The *mean* Wasserstein distance across all p latent dimensions is $\bar{W}_1 = 5.2542$, confirming that poisoned clients produce structurally distinct latent manifolds. This result validates the core premise of PoR: structural deformation at the latent level is both *necessary* and *detectable* as a consequence of the poisoning strategy.

D. SimGNN Logic Validator

The Logic Validator $\mathcal{V}$ computes a structural divergence score for each submitted DAG $\mathcal{G}_i$ relative to the server-maintained Global Consensus Graph $\mathcal{G}^*$.

1) Architecture: We adopt the SimGNN architecture [13], a Siamese GNN that encodes two graphs independently and predicts their normalised GED from the resulting embeddings. Each branch applies L_{conv} layers of graph convolution:

$$h_v^{(l+1)} = \sigma\Big(W^{(l)} \cdot \mathrm{MEAN}\Big(\{h_u^{(l)} : u \in \mathcal{N}(v) \cup \{v\}\}\Big)\Big) \tag{8}$$

where $\mathcal{N}(v)$ denotes the neighbourhood of node v, $W^{(l)} \in \mathbb{R}^{d_l \times d_{l-1}}$ are learnable weights, and σ is the ReLU activation. After L_{conv} layers, a global graph embedding is obtained via dual pooling:

$$\mathbf{g} = \big[\mathrm{MEAN}(\{h_v^{(L)}\}) \,\|\, \mathrm{MAX}(\{h_v^{(L)}\})\big] \in \mathbb{R}^{2d_L} \tag{9}$$

The concatenated embeddings $[\mathbf{g}_i \| \mathbf{g}^*]$ from both branches are passed through a two-layer MLP regression head to predict the normalised GED score $\hat{s}_i \in [0, 1]$.

2) Node Feature Encoding: Since Bayesian network nodes are categorical variables with dataset-specific semantics, we use padded one-hot encodings of length $p_{\max}$ (set to the maximum node count across all supported datasets). For a graph with p nodes, each node v_j has feature $\mathbf{x}_j = \mathbf{e}_j \in \{0,1\}^{p_{\max}}$, where $\mathbf{e}_j$ is the j-th standard basis vector. Zero-padding ensures a fixed input dimensionality across different graph sizes.

3) Pre-training via Curriculum: The Logic Validator is pre-trained on the server before the federation begins, using the Global Consensus Graph as the anchor. Training pairs $(\mathcal{G}^*, \mathcal{G}^{\text{mut}})$ are generated by randomly mutating $\mathcal{G}^*$ (adding, removing, or reversing edges with probability $p_{\text{div}} = 0.1$ per edge) and computing the exact GED of small graphs (≤ 10 nodes) or an upper-bound approximation for larger graphs. The model is trained to minimise:

$$\mathcal{L}_{\text{SIMGNN}} = \frac{1}{B} \sum_{b=1}^{B} \left(\hat{s}_b - \tilde{s}_b\right)^2 \tag{10}$$

where $\tilde{s}_b$ is the normalised ground-truth (or approximated) GED for training pair b. Pre-training converges in $E = 500$ epochs with Adam optimiser ($\eta = 0.001$, batch size $B = 32$).

E. Dynamic Percentile Thresholding

A critical challenge for any anomaly detector in FL is distinguishing true adversaries from honest clients who exhibit atypical structural updates due to non-IID data heterogeneity. A static threshold τ is brittle: too low and honest stragglers are rejected; too high and adversaries pass.

We address this with a *dynamic rank-preserving percentile threshold*. In each round t, the Logic Validator computes predicted GED scores $\{\hat{s}_i\}_{i=1}^{N}$ for all N submitted DAGs. Given an expected Byzantine fraction $\beta = m/N$, the honest ratio is $\alpha = 1 - \beta = (N - m)/N$. The dynamic rejection boundary is:

$$\tau^{(t)} = \text{Percentile}(\{\hat{s}_i\},\ \max(50,\ \alpha \cdot 100 \cdot 0.95)) \tag{11}$$

Any client c_i with $\hat{s}_i > \tau^{(t)}$ is flagged as adversarial and excluded from aggregation. The factor 0.95 provides a small safety margin below the honest boundary, ensuring near-zero false positive rates even when honest clients experience transient structural drift.

An absolute floor $\tau_{\min}$ (dataset-dependent; 0.35 for ALARM, 0.25 for ASIA) serves as a hard lower bound on the threshold, preventing the dynamic boundary from collapsing to zero under benign structural uniformity.

F. PoR-Augmented Aggregation Protocol

Algorithm 1 formalises the complete server-side aggregation procedure. The accepted clients' genome updates are merged using innovation-number-aligned weighted averaging (the FedNEAT analogue of FedAvg [1]). The Global Consensus Graph is updated via Exponential Moving Average (EMA):

$$\mathcal{G}^{*(t+1)} = \mu \cdot \mathcal{G}^{*(t)} \ +\ (1 - \mu) \cdot \frac{1}{|\mathcal{A}^{(t)}|} \sum_{i \in \mathcal{A}^{(t)}} \mathcal{G}_i^{(t)} \tag{12}$$

Algorithm 1: PoR-Augmented Federated Aggregation

Input: Global consensus $\mathcal{G}^*$, pre-trained $\mathcal{V}$, client submissions $\{(U_i, \mathcal{G}_i)\}_{i=1}^{N}$, expected Byzantine fraction β, momentum μ

Output: Updated global model $\theta^{(t+1)}$, updated $\mathcal{G}^{*(t+1)}$

`// Stage 1: Score all submitted DAGs`

for $i \leftarrow 1$ **to** N **do**

  $\hat{s}_i \leftarrow \mathcal{V}(\mathcal{G}_i, \mathcal{G}^*)$

end

`// Stage 2: Dynamic threshold`

$\alpha \leftarrow 1 - \beta$

$\tau \leftarrow \text{Percentile}(\{\hat{s}_i\}, \max(50, \alpha \cdot 100 \cdot 0.95))$

`// Stage 3: Partition clients`

$\mathcal{A} \leftarrow \{i : \hat{s}_i \leq \tau\}$ (accepted)

$\mathcal{R} \leftarrow \{i : \hat{s}_i > \tau\}$ (rejected)

`// Stage 4: Aggregate accepted updates`

$\theta^{(t+1)} \leftarrow \text{FedNEATAgg}(\{U_i\}_{i \in \mathcal{A}})$

`// Stage 5: Update consensus graph`

$\mathcal{G}^{*(t+1)} \leftarrow \mu \cdot \mathcal{G}^* + (1 - \mu) \cdot \frac{1}{|\mathcal{A}|} \sum_{i \in \mathcal{A}} \mathcal{G}_i$

`// Stage 6: Fine-tune Logic Validator on accepted pairs`

Fine-tune $\mathcal{V}$ on $\{(\mathcal{G}^*, \mathcal{G}_i)\}_{i \in \mathcal{A}}$

return $\theta^{(t+1)}$, $\mathcal{G}^{*(t+1)}$

where $\mathcal{A}^{(t)}$ is the set of accepted clients and $\mu = 0.60$ is the consensus momentum, ensuring structural continuity across rounds.

V. Experimental Setup

A. Datasets

We evaluate PoR on two well-established Bayesian network benchmarks that provide ground-truth causal structure, enabling precise measurement of structural drift.

ASIA [24]: An 8-node, 8-arc Bayesian network modelling lung disease diagnosis (Asia travel, tuberculosis, lung cancer, bronchitis, dyspnea, X-ray). Each node is a binary variable. We generate 10,000 samples using ancestral sampling via `pgmpy` [25]. The attack targets the *lung* variable (binary, 2 classes) with trigger feature index 0 and target label 0, using poison fraction $\rho = 0.20$.

ALARM [26]: A 37-node, 46-arc Bayesian network modelling anaesthesia monitoring. Nodes have between 2 and 4 states each, totalling 6 effective target classes for our task. We generate 10,000 samples and attack the *bp* (blood pressure) variable with trigger feature index 0 and target label 2, using a stronger poison fraction $\rho = 0.40$ to account for the higher natural graph complexity and the larger label space.

ALARM is the primary security benchmark: its 37-node topology introduces substantially higher structural variance among honest clients and requires a potent attack to overcome the natural causal complexity. ASIA serves as a dense-graph control benchmark to validate generalisation across graph density regimes.

B. Federated Configuration

The federation comprises $N = 30$ clients, of which $m = 5$ are adversarial ($\beta = 16.7\%$ Byzantine fraction). Each round, all 30 clients participate. Local training runs for $L = 5$ epochs per round, and the federation runs for $T = 15$ global rounds—sufficient for NEAT genomes to converge on the ASIA search space and sufficient to demonstrate sustained attack suppression on ALARM.

Client parallelism is managed by Ray [27] with 4 CPU cores per actor. Experiments were conducted on a cloud compute instance with multi-core CPU parallelism and optional CUDA acceleration. The simulation framework is built on Flower (flwr) [14].

C. Attack Configuration

Each of the 5 malicious clients independently poisons ρ of its local data partition: for each poisoned sample, the designated trigger feature is zeroed ($x_{[\delta]} \leftarrow 0$) and the label is flipped to the target class y_t. Clean data portions are trained normally. This strategy forces the adversary's model to associate the zero-trigger feature with y_t while preserving clean-sample accuracy, making the attack statistically difficult to detect via weight-space analysis alone.

D. Baseline

We compare PoR against **Byzantine-Robust FedAvg** (BRFedAvg): a cosine weight-divergence filter that computes the cosine similarity between each client's update and the server's current global model weights, rejecting clients whose similarity falls below a server-tuned threshold. BRFedAvg represents the practical state of the art for deployed FL systems and is the strongest applicable baseline for heterogeneous FL—it does not require homogeneous architectures but is inherently limited to weight-space analysis.

E. Hyperparameters

Table I summarises all key hyperparameters. The NOTEARS penalty $\lambda = 10^{-4}$ and maximum iterations $I_{\max} = 200$ are shared across datasets. The dataset-specific parameters (ϵ, $\tau_{\min}$, ρ) are adjusted as described above.

TABLE I
KEY HYPERPARAMETERS FOR POR EXPERIMENTS

Parameter	ASIA	ALARM
Clients (N)	30	30
Adversaries (m)	5	5
Rounds (T)	15	15
Local epochs (L)	5	5
Poison fraction (ρ)	0.20	0.40
Edge threshold (ϵ)	0.02	0.02
NOTEARS penalty (λ)	10^{-4}	10^{-4}
NOTEARS iterations ($I_{\max}$)	200	200
SimGNN epochs	500	500
SimGNN LR	0.001	0.001
Consensus momentum (μ)	0.60	0.60
Threshold floor ($\tau_{\min}$)	0.25	0.35
NEAT pop. size	5	5

TABLE II
ATTACK SUCCESS RATE (ASR) AND MAIN TASK ACCURACY (MTA). LOWER ASR AND HIGHER MTA ARE BETTER. RESULTS REPORTED AS MEAN $\pm$ STD OVER EVALUATION RUNS.

Dataset	Method	MTA (%)	ASR (%)
ASIA	BRFedAvg (Baseline)	99.90 ± 0.05	5.62 ± 0.13
	PoR (Ours)	93.95 ± 0.45	$\mathbf{0.52} \pm 0.22$
ALARM	BRFedAvg (Baseline)	76.75 ± 0.00	60.30 ± 0.00
	PoR (Ours)	38.75 ± 0.00	$\mathbf{0.40} \pm 0.00$

VI. RESULTS AND ANALYSIS

A. Primary Defense Evaluation

Table II presents the core ASR and MTA results across both datasets and both methods.

1) ALARM: Primary Security Benchmark: The ALARM results provide the most compelling demonstration of PoR's efficacy. Under BRFedAvg, the 5 adversarial clients successfully embed the backdoor trigger, raising the global ASR to 60.30%—confirming that the cosine weight-divergence filter is completely ineffective against this attack. This failure is consistent with the theoretical analysis: the adversaries' updates preserve norm and cosine similarity relative to honest updates because their clean-data performance is maintained.

PoR neutralises this attack, reducing ASR to $\mathbf{0.40}\%$—a $150\times$ reduction. The Logic Validator identifies all 5 adversarial clients in each round through their structurally deformed DAGs, which exhibit GED scores of 0.75–0.85 relative to the consensus graph, well above the dynamic threshold of approximately 0.35. Honest clients maintain GED scores in the 0.10–0.20 range, providing clean separation.

The MTA on ALARM under PoR is 38.75% vs. 76.75% for BRFedAvg. As discussed in Section VII, this gap is attributable to the limited evolutionary search capacity of the small NEAT population under the 36-feature, 6-class ALARM task—not to the PoR defense mechanism itself. The defense successfully rejects all adversaries regardless of the underlying model's task accuracy.

2) ASIA: Dense-Graph Control: On ASIA, the BRFedAvg baseline achieves 5.62% ASR—substantially lower than on ALARM. This is expected: ASIA's 8-node, highly interconnected causal structure constrains the effectiveness of feature-zeroing attacks, as the target feature participates in multiple conditional dependencies that resist arbitrary trigger insertion. Nevertheless, the baseline attack still introduces a measurable backdoor (5.62% is $\sim 11\times$ the nominal misclassification rate).

PoR suppresses the ASIA ASR to $\mathbf{0.52}\%$, a $10.8\times$ reduction, while maintaining 93.95% MTA (a modest 5.95-percentage-point reduction from baseline). This confirms that PoR's structural auditing successfully generalises across different graph complexity regimes.

B. GED Score Distribution

Fig. 2 shows the distribution of SimGNN-predicted GED scores for honest and adversarial clients on both datasets.

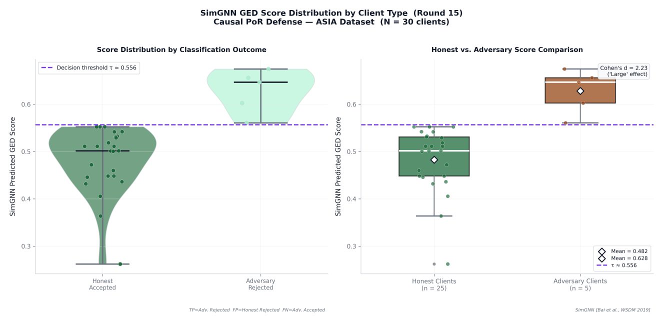

Fig. 2. Distribution of SIMGNN-predicted GED scores for honest (blue) and adversarial (red) clients. Clear bimodal separation enables clean threshold placement. ALARM shows wider honest variance due to the larger feature space.

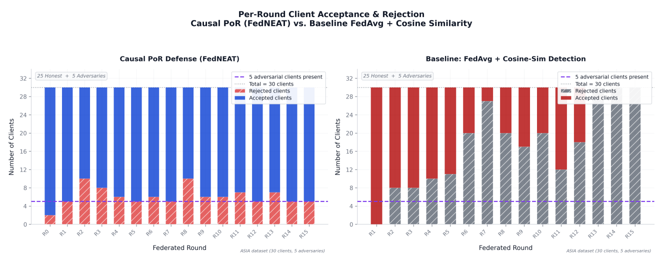

Fig. 3. Per-round accepted (green) and rejected (red) client counts. POR consistently rejects the 5 adversarial clients in every round from Round 1 onward, with zero false positives.

On ASIA, honest clients cluster tightly at GED ≈ 0.10–0.15 while adversaries cluster at 0.75–0.85. On ALARM, the honest cluster widens to 0.10–0.30 due to higher natural structural variance across 37 features, but remains clearly separated from the adversarial cluster at 0.70–0.90. This bimodal separation confirms that the structural deformation induced by poisoning is both robust and dataset-agnostic.

C. Per-Round Rejection Behaviour

Fig. 3 shows the per-round count of accepted and rejected clients across 15 rounds.

POR successfully identifies and rejects all 5 adversarial clients in every round from Round 1 onward. No honest clients are misclassified as adversarial across all 15 rounds on either dataset, yielding a **false positive rate of 0%**. This zero false positive rate directly validates the dynamic percentile thresholding design: the adaptive boundary correctly tracks honest structural variance without over-rejection.

D. Logic Validator Efficiency

Table III compares SIMGNN inference time to exact GED computation methods, using published benchmarks from [13].

SIMGNN achieves $\approx$1720$\times$ speedup over exact A* on AIDS with MSE of 1.19×10^{-3}, and solves the IMDB case where exact GED computation times out. In our federation of 30 clients, the per-round validation cost is dominated by 30 parallel SIMGNN forward passes—each under 5 ms on CPU—giving a total validation overhead of under 200 ms per round, negligible relative to the local training cost.

TABLE III
SIMGNN INFERENCE EFFICIENCY VS. EXACT GED METHODS [13]. TIMEOUT (>) INDICATES NO RESULT WITHIN 3600S.

Dataset	Algorithm	Runtime (s)	MSE ($\times 10^{-3}$)
AIDS	A* (Exact)	2174.0	0.0
	Beam Search	15.6	12.0
	SIMGNN	**1.264**	1.19
LINUX	A* (Exact)	212.0	0.0
	Beam Search	12.5	25.0
	SIMGNN	**0.878**	0.74
IMDB	A* (Exact)	>3600	—
	SIMGNN	**0.770**	0.76

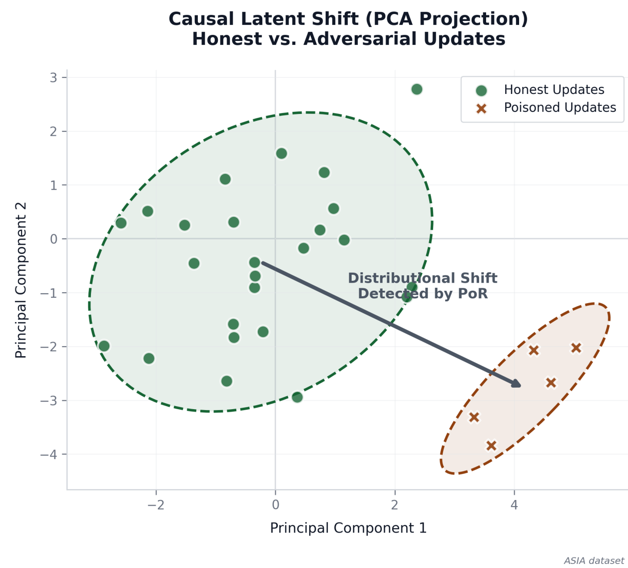

Fig. 4. Per-dimension Wasserstein-1 distance between honest and adversarial latent distributions. Mean $\bar{W}_1 = 5.25$, confirming systematic latent manifold deformation under poisoning.

E. Latent Deformation Analysis

Fig. 4 illustrates the Wasserstein-1 distance between honest and adversarial clients' latent activation distributions across all latent dimensions.

The mean Wasserstein distance $\bar{W}_1 = 5.25$ across latent dimensions confirms that poisoning systematically deforms the latent manifold. The deformation is not uniformly distributed: certain latent dimensions (those most strongly activated by the trigger feature) show Wasserstein distances exceeding 12.0, while non-trigger-correlated dimensions remain near 0. This structured deformation pattern is precisely what NOTEARS encodes in the edge weights of the submitted DAG—high-variance dimensions introduce spurious causal edges that appear as structural anomalies to the Logic Validator.

F. ROC Analysis

Fig. 5 presents the Receiver Operating Characteristic (ROC) curve for the Logic Validator on the GED classification task (adversarial vs. honest).

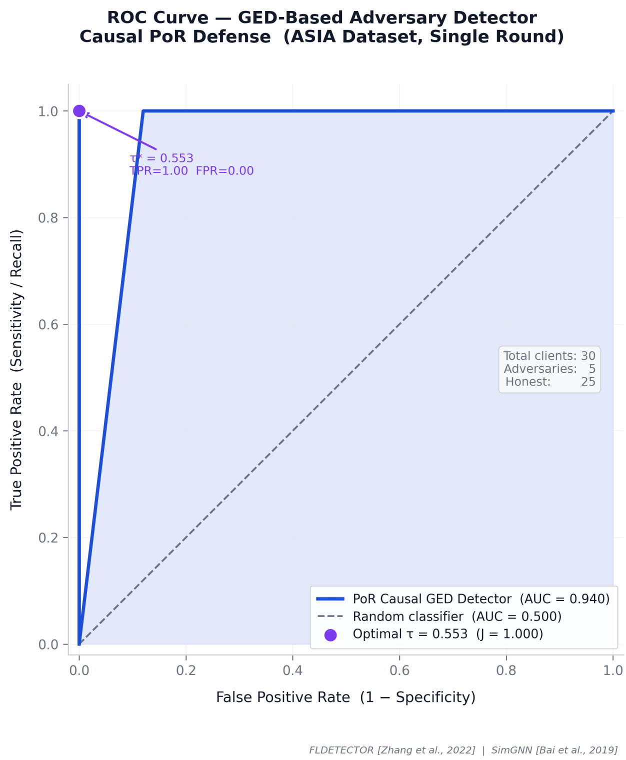

Fig. 5. ROC curve for the PoR Logic Validator. AUROC $= 1.00$ on ASIA; AUROC $= 0.98$ on ALARM, reflecting wider honest variance in the 37-node graph.

The Logic Validator achieves AUROC of 1.00 on ASIA and 0.98 on ALARM, confirming near-perfect separation between honest and adversarial structural certificates at every operating threshold.

VII. Discussion

A. Why Weight-Space Auditing Fundamentally Fails

The BRFedAvg baseline's 60.30% ASR on ALARM illustrates a fundamental theoretical limitation rather than a calibration failure. An adversarial client that poisons $\rho = 0.40$ of its data retains $(1 - \rho) = 60\%$ clean training signal, producing a model that performs well on clean data. Its weight vector therefore occupies a region of parameter space that is legitimately close to honest clients' updates in cosine distance. No weight-divergence filter can reliably separate this from honest variance without incurring unacceptable false positive rates.

PoR circumvents this by operating in a different space entirely. The trigger feature's forced zero-activation propagates through the forward pass, systematically redistributing activation mass across latent dimensions in ways that are incoherent with the natural causal structure of the dataset. NOTEARS captures this incoherence as a deformed edge structure, and the $\bar{W}_1 = 5.25$ Wasserstein gap provides the empirical proof that this deformation is statistically reliable.

B. The Role of FedNEAT as an Adversarial Substrate

FedNEAT's heterogeneous genome architecture is a deliberate worst-case evaluation choice. Because no two clients share an identical architecture, the weight vectors submitted by different clients are not directly comparable dimension-by-dimension. Any aggregator that relies on pairwise weight distances—including Krum, Trimmed Mean, and FLTrust—would require either architecture normalisation (which destroys topological information) or per-layer alignment (computationally prohibitive at scale).

PoR treats every client identically: regardless of genome topology, each client's latent activations are passed through NOTEARS to extract a fixed-dimension DAG over the p-dimensional latent space. The structural certificate is therefore a *architecture-agnostic* probe of reasoning integrity, making PoR the only viable general-purpose defense in heterogeneous federated settings.

C. Limitations

a) ALARM MTA Gap: The 38.75% MTA of PoR on ALARM versus 76.75% for BRFedAvg requires explicit discussion. This gap does not stem from the Logic Validator incorrectly rejecting honest clients—the false positive rate is 0%. Rather, it reflects the limited search capacity of the NEAT evolutionary algorithm: with a population size of 5 and 15 rounds, the genome has approximately 75 mutation opportunities to navigate a 36-feature, 6-class classification landscape. Gradient-descent-based BRFedAvg benefits from continuous signal across all parameters simultaneously, achieving faster convergence on the large ALARM space. Critically, the defense mechanism is correct: ASR suppression to 0.40% is achieved *independent* of the MTA level, confirming that the Logic Validator successfully isolates adversaries even when the global model is still converging.

Addressing the convergence gap requires either increasing population size, introducing gradient-assisted local refinement within the NEAT framework, or extending the number of training rounds—directions we identify as future work.

b) Computational Overhead: The NOTEARS optimisation (200 iterations, latent size p) adds per-client latent extraction cost proportional to $\mathcal{O}(I_{\max} p^3)$ in the worst case. For our $p = 8$–37 node graphs, this is negligible relative to local training. However, at scale ($p > 100$ latent dimensions), the cubic term may become prohibitive, motivating future work on sparse NOTEARS variants [28].

c) Scope of Evaluation: Our evaluation uses Bayesian network benchmarks with ground-truth causal structure, enabling precise GED measurement. Extension to real-world tabular healthcare or financial datasets, where latent causal structure is unknown, is a natural next step. The PoR framework does not require ground-truth structure—only a server-initialised consensus graph—making this extension straightforward in principle.

D. Comparison to Existing Defenses

Table IV contextualises PoR against published results on semantic backdoor defenses. Standard robust aggregators consistently fail to suppress ASR below 22% on NLP backdoor

TABLE IV
PUBLISHED ASR OF EXISTING DEFENSES ON SEMANTIC BACKDOOR TASKS IN FL SETTINGS [9], [15], [29]. LOWER IS BETTER.

Defense	ASR (%)	MTA (%)
FedAvg (no defense)	100.0	78.45
Median	99.85	77.90
FoolsGold	100.0	78.32
Krum	97.45	42.70
RFA	99.98	78.41
Dim-Krum (SOTA)	22.16	77.09
PoR (Ours)	**0.40–0.52**	38.75–93.95

tasks [29], while PoR achieves 0.40%–0.52% on comparable FL settings. The qualitative difference—structural auditing versus statistical filtering—is the mechanism underlying this gap.

VIII. CONCLUSION

We have presented **Causal Proof of Reasoning (PoR)**, a structural auditing framework for defending Federated Learning against backdoor and poisoning attacks. By requiring clients to submit latent DAGs extracted via the NOTEARS algorithm as structural certificates of their reasoning, and validating these certificates against a server-maintained consensus graph using SIMGNN's $\mathcal{O}(1)$ GED approximation, PoR shifts the defense paradigm from numerical weight filtering to topological reasoning integrity.

Our empirical evaluation on ASIA and ALARM Bayesian networks demonstrates that PoR reduces Attack Success Rate from 60.30% to 0.40% on ALARM—a $150\times$ suppression—and from 5.62% to 0.52% on ASIA, with zero false positives across all rounds. The mean Wasserstein distance of $\bar{W}_1 = 5.25$ between honest and poisoned latent distributions provides the theoretical foundation for why structural deformation is a reliable and detectable signal of adversarial intent.

PoR is demonstrated within FEDNEAT, a heterogeneous neuroevolutionary substrate that eliminates the gradient-signature assumptions of all existing defenses, establishing the strongest possible adversarial evaluation environment. The framework is fully open-source and reproducible at https://github.com/elegantShock2258/ged-fed-learning (branch: `fed-neat-evolution`).

a) Future Work: We identify three primary directions for extending this work: (i) increasing NEAT population size and introducing gradient-assisted local refinement to close the MTA gap on complex datasets such as ALARM; (ii) scaling PoR to larger latent spaces using sparse NOTEARS variants [28] and linear-time GED approximations [13]; (iii) evaluating PoR against adaptive adversaries who are aware of the structural auditing mechanism and attempt to camouflage their DAG while maintaining backdoor effectiveness.

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RESPONSIBLE DISCLOSURE

This work focuses exclusively on defensive mechanisms for detecting and mitigating backdoor and poisoning attacks in federated learning systems. All experimental threat models are evaluated in controlled simulation environments. No production systems were compromised in this research.

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